Name: Omar Rakhe Batch: T22 Roll No. 94

# Experiment 10

**Aim:** Learn Dockerfile Instruction and Build Image

**Step 1: Downloading Docker**

The first place to start is the **official Docker website** from where we can download Docker Desktop.

**Step 2: Configuration**

To run Linux on Windows, Docker requires a virtualization engine. Docker recommends using WSL 2

**Step 3: Running the installation.**

Click Ok, and wait a bit…

**Step 4: Restart**

For Docker to be able to properly register with Windows, a restart is required at this point.

**Step 5: WSL 2 installation**

After you accept the license terms, the Docker Desktop window will open. However, we are not done yet. Since we have selected WSL 2 as our virtualization engine, we also need to install it. **Don't click Restart just yet!**

**Step 6— Starting Docker Desktop**

If Docker Desktop did not start on its own, simply open it from the shortcut on your Desktop. If you wish, you can do the initial orientation by clicking Start.

**Step 7— Testing Docker**

Open your favourite command line tool and type in the following command: docker run hello-world.

This will download the hello-world Docker image and run it. This is just a quick test to ensure everything is working fine.

**Step 8 — Automatically start Docker [Optional]**

**This step is dependently optional**, but if you are working a lot with Docker, you may want to configure Docker Desktop to start automatically once you start Windows.

Troubleshoot — Issues installing WSL 2 Step

## 9 — Enable Virtual Machine Platform

Make sure that Virtual Machine Platform is enabled on your Windows installation. You can easily check this by opening the *Turn Windows features on or off* from your Control panel.

```
C:\Users\Lenovo>docker run redis
1:C 24 Feb 2026 07:57:09.266 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 24 Feb 2026 07:57:09.266 * Redis version=7.2.4, bits=64, commit=00000000, modified=0, pid=1, just started
1:C 24 Feb 2026 07:57:09.266 # Warning: no config file specified, using the default config. In order to specify a config
 file use redis-server /path/to/redis.conf
1:M 24 Feb 2026 07:57:09.266 * monotonic clock: POSIX clock_gettime
1:M 24 Feb 2026 07:57:09.267 * Running mode=standalone, port=6379.
1:M 24 Feb 2026 07:57:09.268 * Server initialized
1:M 24 Feb 2026 07:57:09.269 * Ready to accept connections tcp
```

```
C:\Users\Lenovo>docker pull redis
Using default tag: latest
latest: Pulling from library/redis
0c8d55a45c0d: Pull complete
9a00fba7918f: Pull complete
253eb0a88de9: Pull complete
2ccab22a2cd8: Pull complete
5a312569f2e0: Pull complete
4f4fb700ef54: Pull complete
fb0cbf9357d9: Pull complete
Digest: sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview redis
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE                                    COMMAND                  CREATED         STATUS          PORTS
      NAMES
1c17503a759c   170a1e90f843                             "docker-entrypoint.s…"   5 minutes ago   Up 22 seconds   6379/
tcp   amazing_robinson
a388ed3e2ae7   docker/dev-environments-default:stable-1 "sleep infinity"         23 months ago   Up 26 seconds
      siddhesh-app-1
```

```
C:\Users\Lenovo>docker stop 1c17503a759c
1c17503a759c

C:\Users\Lenovo>docker start 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker rm 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker images
REPOSITORY                      TAG                                       IMAGE ID       CREATED         SIZE
redis                           latest                                    f0acdeb5ced3   14 hours ago    140MB
ubuntu                          latest                                    802541663949   6 months ago    78.1MB
ubuntu                          <none>                                    ca2b0f26964c   24 months ago   77.9MB
redis                           <none>                                    170a1e90f843   2 years ago     138MB
docker/dev-environments-default stable-1                                  7eb85f44c229   3 years ago     610MB
crazymax/linguist               7.20.0                                    dc5f9ae3be6a   3 years ago     72.2MB
docker/desktop-git-helper       5a4fca126aadcd3f6cc3a011aa991de982ae7000  efe2d67c403b   4 years ago     44.2MB
```